

Technology Stack Breakdown

Core Technology Layers

ShopEEZ uses MERN stack architecture. The database layer uses Mongoose with Atlas, Node.js and Express manage API routes, and React 19 provides the user interface.

Stack Layer	Selected Technology	Role in ShopEEZ Platform	Libraries / Extensions
Frontend Framework	React 19 (Vite.js)	Component-based SPA structure for rapid page routing.	React Router DOM, Redux Toolkit
State Store	Redux Toolkit	Global client state management (Auth session, Cart list, Wishlist).	Redux Persist wrapper
CSS Styling	Vanilla CSS	Responsive styling grids and cards matching local device layout.	Flexbox, CSS variables
Server Runtime	Node.js (v18+)	Underlying JavaScript execution engine for backend services.	NPM packages manager
API Middleware	Express.js	REST controller route handling, security rules, image processing.	jsonwebtoken, bcryptjs, multer
Database Store	MongoDB Atlas	Document database for saving client accounts, catalogs, orders.	Mongoose ODM library



